Title : Secure Coding Review Report

Task : CodeAlpha Cyber Security – Task 3

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INTRODUCTION :

This report presents a secure coding review of a simple Python login application.
The goal of this review is to identify security vulnerabilities in the code and
suggest best practices for improving application security.

Code Overview :

The reviewed application is a basic login system written in Python.

It takes a username and password from the user and checks them using a simple if-condition.

CODE :

```
username = input("Enter username: ")
password = input("Enter password: ")


if username == "admin" and password == "1234":
    print("Login successful")
else:
    print("Access denied")
```

Vulnerabilities :

1. **Hardcoded Password**

   The password is written directly in the code, making it easy to compromise.

2. **Plain Text Password**

   Passwords are not encrypted or hashed.

3. **No Input Validation**

   The program does not sanitize user input.

4. **No Login Attempt Limit**

   Unlimited attempts allow brute-force attacks.

Recommendations & Best Practices :

Recommended Fixes

1. Use hashed passwords (e.g., SHA-256).
2. Validate all user inputs.
3. Implement proper authentication systems.
4. Limit login attempts.
5. Use secure communication (HTTPS).

Remediation Steps :

To improve security, the application should store passwords using hashing, validate user input, and use a secure authentication mechanism.
Adding login attempt limits will help prevent brute-force attacks.

Conclusion:

Secure coding is essential for protecting applications from cyber threats.
By fixing the identified vulnerabilities, the login system can become safer and more reliable.